Science Olympiad
Machines B
Purdue Invitational

January 31, 2026

Directions:

- Each team will be given **50** minutes to complete this test.
- This test can be taken apart, but should not be written on **at all**. All questions should be answered on the provided answer sheet.
- There are three sections: **A** (Multiple Choice Questions), **B** (Short Answer Questions), and **C** (Free Response Questions). These sections can be completed in any order.
- Use $g = 9.81 \text{ m/s}^2$. Questions with pi can be left with π in the answer. If the numerical value of pi must be used, use $\pi = 3.14$.
- Appropriate units and significant figures should be provided with all answers.
- Images provided are not necessarily to scale.
- Good luck!

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Section A: Multiple Choice Questions (30 points)

All questions in this section are worth 2 points each. Only the letters 'A' through 'D' should be written on the answer sheet. 'T' and 'F' will NOT be accepted.

1. The coefficient of kinetic friction μ_k is generally smaller than the coefficient of static friction μ_s.

   (a) True (b) False

2. An inelastic collision conserves momentum.

   (a) True (b) False

3. Mechanical advantage relates the input and output forces.

   (a) True (b) False

4. Friction is a conservative force.

   (a) True (b) False

5. Efficiency of an ideal machine is always 1.

   (a) True (b) False

6. Which of the following is an example of a class 2 lever?

   (a) Crowbar
   (b) See-saw
   (c) Tweezers
   (d) Wheelbarrow

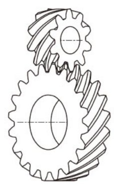
7. Which of the following is NOT true for an ideal machine?

   (a) Input work must equal output work
   (b) Energy can be lost to irreversibilities such as friction or heat
   (c) The machine has to violate the second law of thermodynamics
   (d) Efficiency must be equal to 100 %

8. When a skydiver reaches terminal velocity, which of the following values is NOT constant?

   (a) Kinetic energy
   (b) Total mechanical energy
   (c) Momentum
   (d) Acceleration

9. Which type of gear is pictured in the figure below?



   (a) Rack and pinion
   (b) Miter gear
   (c) Screw gear
   (d) Worm gear

10. Which of the following actions is LEAST likely to increase the efficiency of a machine?

   (a) Lubricating parts where surface-to-surface contact occurs
   (b) Using thermal insulation on the boundary of the system
   (c) Increasing the power input of the machine
   (d) Adding ball bearings to joints

11. Which object has the most gravitational potential energy?

   (a) An apple (0.15 kg) held at the top of the Empire State Building (381 m)
   (b) A skier with mass 60kg travelling at 40 km/h and total mechanical energy 4 kJ
   (c) You! Currently. Provided that your feet are on the floor
   (d) A dog (30 kg) at the apex of a parabolic jump satisfying $y = -x^2 + 1.6x$

12. What is the efficiency of a compound machine with two simple machines with efficiencies e_1 and e_2?

   (a) $e_1 + e_2$
   (b) $e_1 e_2$
   (c) $\dfrac{e_1 + e_2}{e_1 e_2}$
   (d) $\dfrac{e_1 e_2}{e_1 + e_2}$

13. When driving a screw into a wall, what is the relationship between the force applied to the screwdriver and the force exerted by the screw on the wood?
    (a) The applied force is greater than the force exerted by the screw
    (b) The applied force is less than the force exerted by the screw
    (c) The applied force is exactly equal to the force exerted by the screw
    (d) There is no consistent relationship between the two forces
14. Which of the following have the same SI units?
    (a) Power and impulse
    (b) Energy and momentum
    (c) Acceleration and drag
    (d) Torque and energy
15. Which of the following is true for a rack and pinion?
    (a) The rack moves in a circular motion when driven by the pinion
    (b) When the rotational element has constant angular velocity, the flat element accelerates
    (c) It turns rotational motion into linear motion
    (d) It turns mechanical energy into electrical energy

Section B: Short Answer Questions (32 points)

All questions in this section are worth 4 points. The following questions will require a single numerical answer. There is no partial credit.

1. Consider 2 gears. How many teeth does the input gear have if the output gear has 36 teeth and the IMA of the system is 4?
2. A machine has an ideal mechanical advantage of 9 and an efficiency of 0.6. What is the actual mechanical advantage?
3. Consider a car with mass 2000kg driving down the freeway at 120 km/h. What is the translational kinetic energy of the car in MJ?
4. The engine in an electric bike operates at an ideal power output of 480W. Unfortunately, it has an efficiency of 67 %. What is the difference between the theoretical and actual time that the bike will take to output 100 kJ?
5. A wedge has the shape of an isoceles triangle with a base of length 4 cm and with side lengths of 8 cm. What is the IMA of this wedge?
6. Some machine has a power input of 8 kW. However, energy is lost in the form of heat, due to friction, at a rate of 1000 joules per second. What is the efficiency of this machine?
7. Calculate the mechanical advantage of a road that climbs from sea level to a height of 3000 feet over 8 miles.
8. What is the IMA of an M2 screw with a pitch of 0.4mm and a diameter of 2mm?

Section C: Free Response Questions (38 points)

Point values are denoted on the individual questions. *(Questions will likely require multiple sentences and/or work shown to succesfully earn all points.)*

1. [19 pts] Consider the following situation that is pictured below in Figure 1. There is a ramp of height 2 meters at an angle of 24 degrees relative to the ground. There is a metal block that has a mass of 5 kg held at rest at the top of the ramp. It is given that the ground has a coefficient of friction of 0.3 and the ramp is frictionless. If the block is released, what is the

   (a) [3 pts] velocity of the block as it leaves the ramp?

   (b) [4 pts] total distance travelled by the block from the base of the ramp?

   Now assume that the ramp is also rough.

   (c) [6 pts] What is the smallest that the coefficient of static friction can be for the block to not move?

   (d) [6 pts] Imagine that the block does move after release and travels along the ramp and then stops 5 meters from the base of the ramp. What is the coefficient of kinetic friction of the ramp? Remember that the ground still has a coefficient of kinetic friction of 0.3.

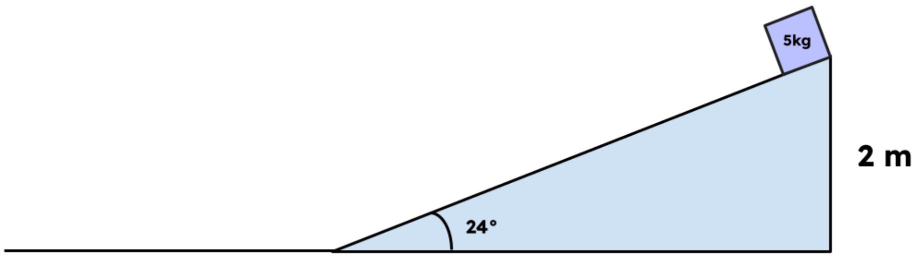


Figure 1

2. [19 pts] Consider the ideal pulley pictured in Figure 2 with frictionless connections and massless wheels and strings. Mass 2 is held 0.5 meters above the ground. If $m_1 = 3$ kg and $m_2 = 8$ kg,

   (a) [3 pts] determine the acceleration of the two masses when released from rest.

   Now imagine that the mass 2 travels straight down and elastically transfers its energy to the ideal lever positioned directly below. This energy is then elastically transferred to mass 3, causing it to be launched upward. Rotational effects and air resistance are ignored and it is assumed that the balls will only travel vertically.

   (b) [3 pts] How long will it take for mass 2 to hit the ground?

   (c) [5 pts] How far will a bowling ball with mass 6 kg travel? What about a bouncy ball with mass 0.2 kg?

   My buddy Zorp from Krootabulon-A is messing with the three body system in Figure 3. The tension in the main string is equal to T, and $m_1 = M, m_2 = 2M$, and $m_3 = 3M$. The left and right pulleys are fixed, while the center pulley can translate vertically. Assume that the pulleys are ideal and the strings are massless. Note the gravity on Krootabulon-A is NOT equivalent to that of Earth. Instead, it is equal to some constant g_k.

   (d) [3 pts] Draw free body diagrams for each of the three masses.

   (e) [4 pts] Find a symbolic expression for g_k, in terms of T and M.

   (f) [1 pts] If $T = 36$ N and the mass M is equal to 5 kg, find the gravitational acceleration on Krootabulon-A, g_k.

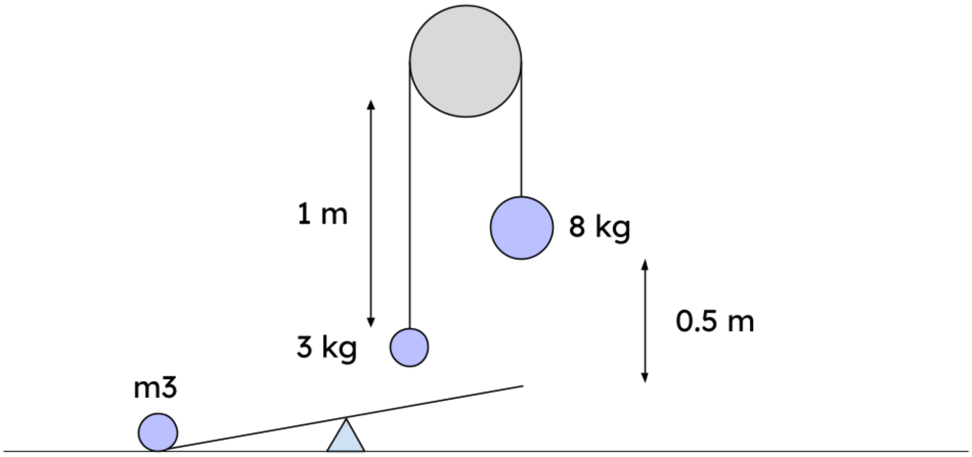


Figure 2

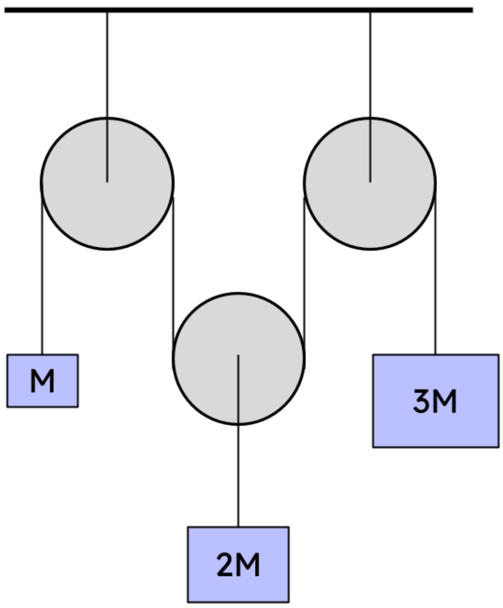


Figure 3